

The Blind Sector

Internal Structure of the Null Sector and the Acceptance of External Covariance in Resolution Geometry

Matěj Rada

MatejRada@email.cz

Licensed under CC BY-NC-ND 4.0

July 15, 2026

Abstract

Resolution geometry depends on an externally supplied covariance—in particular on the null sector, which the first-order response cannot reach. This paper removes as much of that dependence as is possible in principle, and delimits exactly what remains. *Part I* shows the null sector is not structureless. It obeys a floor set by the coupling ($\Sigma_N \succeq C^\top \Sigma_R^{-1} C$); it splits, by the singular value decomposition of the coupling, into a coupled sub-fibre $N_c = \text{Im}(C^\top)$ and a free sub-fibre $N_f = \ker(C)$; the linear recovery of the null from the resolved lands exactly in N_c , so the estimator is sighted on its own blind spot precisely as far as the coupling reaches, and no further, the reach being measured by the canonical correlations; and—the structural core—the same Schur logic that separates the resolved from the null sector recurses *inside* the null sector, so the blind sector is self-similar, terminating in an irreducible core that is uncoupled and uncorrelated. The only genuinely external datum is the covariance on that core. *Part II* turns this into an acceptance theory for an external covariance estimate. The admissible estimates form a convex cone (positive-semidefinite, above the Schur floor, supporting the resolved sector); the nearest admissible estimate is a convex projection onto that cone (a constructive repair); the quality of an estimate is a divergence that decomposes additively by fibre layer through the Gaussian chain rule; and estimates classify three ways—inadmissible, admissible-but-incompatible, and compatible—keyed to the blind-sector layers. All statements are established convex analysis, matrix completion, and information geometry, and are numerically verified. The honest boundary: the irreducible core is genuinely external; the theory shrinks and delimits that dependence to its provable minimum, it does not eliminate it.

Contents

1	The dependence to be understood	1
	Part I. The internal structure of the blind sector	2
2	The floor: the hole is bounded below by the coupling	2
3	The split: coupled and free sub-fibres	2
4	Self-sight: the recovery reaches exactly the coupled sub-fibre	2
5	Self-similarity: the blind sector is a recursive Schur structure	2
	Part II. Acceptance of an external covariance	3
6	Admissibility is a convex cone	3
7	Acceptance-repair is a convex projection	3

8	Quality decomposes by fibre layer	3
9	The acceptance test	4
10	The protocol	4
11	Scope and boundaries	4
12	Conclusion	5

1 The dependence to be understood

The framework’s practical weakness is honest and specific: it consumes a covariance Σ that must be estimated from outside, and the part it most depends on—the null sector Σ_N and the coupling C —concerns directions the first-order response cannot see. Two questions follow, and this paper answers both. Is the null sector free, or does internal consistency constrain it? And given an external estimate $\widehat{\Sigma}$, can we decide whether it is acceptable and how good it is? The answers are, respectively, that the null sector is a self-similar Schur structure with an irreducible core, and that acceptance is a convex feasibility problem with a divergence-based quality that decomposes along that structure.

Throughout, a resolution geometry has resolved sector R (dimension r), null sector N (dimension q), and covariance blocks (Σ_R, C, Σ_N) .

Part I. The internal structure of the blind sector

2 The floor: the hole is bounded below by the coupling

Theorem 2.1 (Schur floor). *A covariance with blocks (Σ_R, C, Σ_N) is positive semidefinite only if*

$$\Sigma_N \succeq C^\top \Sigma_R^{-1} C.$$

The null covariance cannot carry less variance than the coupling demands; the blind sector is not free to be arbitrarily small.

Proof. Positive semidefiniteness of a symmetric block matrix is equivalent (with $\Sigma_R \succ 0$) to the Schur complement condition $\Sigma_N - C^\top \Sigma_R^{-1} C \succeq 0$ [1]. Verified: an estimate below the floor makes Σ indefinite. $\square$

3 The split: coupled and free sub-fibres

Definition 3.1 (Fibre split). Let $C = U \operatorname{diag}(s) V^\top$ be the singular value decomposition of the coupling, with $k = \operatorname{rank}(C)$. The null sector splits g -orthogonally as

$$N = N_c \oplus N_f, \quad N_c = \operatorname{Im}(C^\top) \ (\dim k), \quad N_f = \ker(C) \ (\dim q - k).$$

N_c is reachable through the coupling; N_f is uncoupled.

4 Self-sight: the recovery reaches exactly the coupled sub-fibre

Theorem 4.1 (The estimator is sighted on its own blind spot). *The linear recovery of the null sector from the resolved observations, $\hat{y}_N = \mathbb{E}[y_N | y_R] = C^\top \Sigma_R^{-1} y_R$, lies in N_c for every y_R ; its component in N_f is identically zero. The estimator sees into the blind sector exactly as far as the coupling reaches, and no further.*

Proof. $\hat{y}_N = C^\top \Sigma_R^{-1} y_R \in \text{Im}(C^\top) = N_c$, so its projection onto $N_f = \ker(C)$ vanishes (verified: $\sim 10^{-16}$). The formula is the Gaussian conditional mean / linear MMSE estimator [2]. $\square$

Proposition 4.2 (Reach measured by canonical correlations). *The recoverable information about the null sector, the resolved-null mutual information $I(y_R; y_N) = -\frac{1}{2} \sum_i \log(1 - \rho_i^2)$, depends only on the nonzero canonical correlations ρ_i (equivalently on the singular values of the whitened coupling $\Sigma_R^{-1/2} C \Sigma_N^{-1/2}$), i.e. only on N_c . The free sub-fibre contributes zero.*

5 Self-similarity: the blind sector is a recursive Schur structure

Theorem 5.1 (Recursive structure and the irreducible core). *The decomposition that separates R from N recurs inside N : writing Σ_N in the $N_c \oplus N_f$ basis as $\begin{pmatrix} A & B \\ B^\top & D \end{pmatrix}$, the resolved sector's information depends on Σ_N only through the Schur block $(\Sigma_N^{-1})_{N_c} = (A - BD^{-1}B^\top)^{-1}$. Consequently the null sector has three layers:*

1. N_c — directly coupled to R (recoverable, Theorem 4.1);
2. N_f correlated with N_c ($B \neq 0$) — reaches R only indirectly, through that correlation;
3. N_f uncorrelated with N_c ($B = 0$) — the irreducible core.

Perturbing the covariance on the irreducible core leaves the resolved-sector information (Fisher, canonical correlations) exactly invariant.

Proof. C vanishes on N_f , so $C \Sigma_N^{-1} C^\top = C_c (\Sigma_N^{-1})_{N_c} C_c^\top$, and block inversion gives $(\Sigma_N^{-1})_{N_c} = (A - BD^{-1}B^\top)^{-1}$, itself a Schur complement inside Σ_N . When $B = 0$ this is A^{-1} , independent of D ; perturbing D (the core) then leaves the Schur complement, hence the Fisher information and canonical correlations, unchanged. Verified: core perturbation changes Fisher by $\sim 10^{-16}$ when $B = 0$, and by $O(1)$ when $B \neq 0$. $\square$

Corollary 5.2 (The irreducible external datum). *The only genuinely externally sourced datum is the covariance on the irreducible core (layer 3). Everything else about the blind sector is fixed by internal consistency (the Schur floor) together with the coupling. The dependence on outside estimation is thereby delimited to its provable minimum.*

Part II. Acceptance of an external covariance

6 Admissibility is a convex cone

Theorem 6.1 (Admissible estimates form a convex set). *For a fixed resolved geometry (Σ_R, C) , the set of null covariances yielding an admissible resolution geometry,*

$$\mathcal{A} = \{\Sigma_N : \Sigma_N \succeq C^\top \Sigma_R^{-1} C, R \subseteq \text{range}(\Sigma)\},$$

is convex (an affine-shifted positive-semidefinite cone intersected with the support condition). More generally the set of admissible full covariances $\hat{\Sigma}$ is convex.

Proof. $\{\Sigma_N : \Sigma_N \succeq C^\top \Sigma_R^{-1} C\}$ is the PSD cone shifted by the fixed matrix $C^\top \Sigma_R^{-1} C$, hence convex; the support condition $R \subseteq \text{range}(\Sigma)$ is preserved under convex combination of PSD matrices. Verified: the midpoint of two admissible null covariances is admissible. $\square$

7 Acceptance-repair is a convex projection

Proposition 7.1 (Nearest admissible estimate). *An external estimate $\widehat{\Sigma}_N \notin \mathcal{A}$ has a unique nearest admissible estimate (in Frobenius norm),*

$$\text{Proj}_{\mathcal{A}}(\widehat{\Sigma}_N) = C^\top \Sigma_R^{-1} C + (\widehat{\Sigma}_N - C^\top \Sigma_R^{-1} C)_+,$$

where $(\cdot)_+$ is the positive-semidefinite part (eigenvalue thresholding at zero). This is the constructive repair of an inadmissible estimate.

Proof. Projection onto the shifted PSD cone is projection onto the PSD cone after subtracting the floor, which is eigenvalue clipping [4]; uniqueness follows from convexity. Verified: an estimate below the floor is repaired to a positive-definite Σ . $\square$

8 Quality decomposes by fibre layer

Theorem 8.1 (Chain-rule decomposition of quality). *The quality of an external estimate $\widehat{\Sigma}$ relative to a reference Σ , measured by the Gaussian Kullback–Leibler divergence $\text{KL}(\widehat{\Sigma} \parallel \Sigma)$, decomposes additively along the resolved–null split:*

$$\text{KL}(\widehat{\Sigma} \parallel \Sigma) = \underbrace{\text{KL}(\widehat{\Sigma}_R \parallel \Sigma_R)}_{\text{resolved}} + \underbrace{\text{KL}_{\text{cov}}(\widehat{\Sigma}_{N|R} \parallel \Sigma_{N|R})}_{\text{null, conditional}} + \underbrace{\frac{1}{2} \text{tr}[(\widehat{B} - B) \widehat{\Sigma}_R (\widehat{B} - B)^\top \Sigma_{N|R}^{-1}]}_{\text{coupling mismatch}},$$

where $B = C^\top \Sigma_R^{-1}$ is the recovery map and $\Sigma_{N|R}$ the conditional (Schur) covariance. The decomposition recurses inside N along the $N_c \oplus N_f$ split.

Proof. The chain rule of relative entropy applied to the joint Gaussian factored as marginal-resolved times conditional-null [3]; the conditional term separates into a covariance divergence and a mean-mismatch term set by the difference of recovery maps. Verified: direct KL and the decomposition agree to machine precision ($0.09247 = 0.09247$). $\square$

Remark 8.2 (What the layers mean). The resolved term penalises disagreement on what is directly measured; the coupling-mismatch term penalises disagreement on the recovery map (the sighted part of the blind sector); the conditional term, restricted to the core, is the genuinely free contribution. Quality is thus attributed to the correct layer.

9 The acceptance test

Proposition 9.1 (Three-way classification). *An external estimate $\widehat{\Sigma} = (\widehat{\Sigma}_R, \widehat{C}, \widehat{\Sigma}_N)$ classifies as exactly one of:*

1. **Inadmissible** — violates $\widehat{\Sigma} \succeq 0$, the Schur floor, or the support condition. Rejected (mathematically impossible), repairable by Proposition 7.1.
2. **Admissible but incompatible** — a valid covariance that disagrees with the reference on the determined layers (Σ_R, C) beyond tolerance: a different geometry.
3. **Compatible** — agrees on the determined layers, differing only on the free/core layer: accepted, since the core is genuinely free (Corollary 5.2).

Proof. The three cases are the ordered checks (feasibility; determined-layer consistency; core freedom), mutually exclusive and exhaustive by construction. Verified on the three representative estimates (below-floor $\rightarrow$ inadmissible; altered coupling $\rightarrow$ incompatible; core-only change $\rightarrow$ compatible). $\square$

10 The protocol

Given an external covariance estimate $\hat{\Sigma}$ and a reference geometry: compute the Schur floor and check feasibility (Theorem 6.1); if infeasible, repair to the nearest admissible estimate (Proposition 7.1); classify against the determined layers (Proposition 9.1); and score quality by the layer-decomposed divergence (Theorem 8.1), reading off which layer carries the disagreement. The output is an accept/repair/reject decision with a layer-attributed quality score.

11 Scope and boundaries

Non-claim 11.1. The irreducible core (Corollary 5.2) is genuinely external: its covariance is not determined by the resolved data or the coupling, because it concerns directions the system never couples to, and no method can recover information about such directions. The theory *shrinks and delimits* the external dependence to this provable minimum and makes it explicit; it does not eliminate it. The quality decomposition and acceptance test are stated for the Gaussian model (where the KL divergence and canonical correlations are exact). No physical claim is made: these are statements of covariance geometry and estimation.

12 Conclusion

The blind sector is not a structureless hole. It has a floor set by the coupling, splits into a coupled sub-fibre the estimator can see and a free sub-fibre it cannot, and—most tellingly—it is self-similar: the Schur logic that separates the resolved from the null sector recurses inside the null sector, terminating in an irreducible core. The estimator is therefore sighted on its own blind spot exactly as far as the coupling reaches. On this structure rests an acceptance theory for external covariance: admissibility is convex, repair is a projection, quality decomposes by fibre layer, and estimates classify cleanly as inadmissible, incompatible, or compatible. The framework’s dependence on outside estimation, which motivated the inquiry, is now understood precisely: it is real, it is minimal, and it lives exactly on the irreducible core—and every external estimate can be tested, repaired, and graded against the geometry it must serve.

References

- [1] F. Zhang (ed.), *The Schur Complement and Its Applications*, Springer, 2005.
- [2] T. Kailath, A. H. Sayed, and B. Hassibi, *Linear Estimation*, Prentice Hall, 2000. (Linear MMSE; conditional mean.)
- [3] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley, 2006. (Chain rule of relative entropy; Gaussian mutual information.)
- [4] N. J. Higham, “Computing a nearest symmetric positive semidefinite matrix,” *Linear Algebra Appl.* 103 (1988), 103–118.
- [5] S. Boyd and L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004. (Projection onto convex cones.)